

(1)

Chapter: 1

Problems: Set A

P1.1A(a)

∴ Tabular Analysis:-

Trans- actions	Assets				= Liabilities + Owner's Equity			
	Cash	Accounts Receivable	Supplies	Equipment	Accounts Payable	Owner's Capital	Owner's Drawing	Revenue - Expense
1.	+15000					+15000		
2.	-600							+600
3.	-3000			+3000				
4.					+700			+700
5.	-900		+900					
6.	+3000	+7000						
7.	-600						+600	
8.	-500				-500			
9.	-2500							+2500
10.	+4000	-4000						
	$\$13,900 + \$3000 + \$900 + \$3000 =$				$\$200 + \$15000 - \$600 + \$10000 - \$3800$			
	$\$20,800$				$\$20,800$			

$$\therefore \text{Assets} = \$13,900 + 3000 + 900 + 3000 = \$20,800$$

$$\therefore \text{Liabilities + Owner's Equity} = 200 + 15000 - 600 + 10000 - 3800 = \$20,800$$

$$\therefore \text{Assets} = \text{Liabilities + Owner's Equity}$$

(b)

$$\therefore \text{Net Income} = \text{Revenues} - \text{Expenses} = 10,000 - 3,800$$

$$= \$6,200 \text{ (Answer)}$$

(2)

(1)P1.2A(a)

∴ Tabular Analysis:-

	Assets				=	Liabilities		+	Owner's Equity			
Trans- actions	Cash	+ Accounts Receivable	+ Supplies	+ Equipment	=	Notes Payable	+ Accounts Payable	+	Owner's Capital	Owner's Draw- ing	Reve- + nue	Exp- - erte
Balance (July 31)	5000	1500	500	6000	=	0	4200		8800	0	0	0
1.	+1200	-1200										
2.	-2800						-2800					
3.	+4000	+3500										
4.	-400			+2000			+1600				+7500	
5.	-4100											+4100
6.	-700									↓700		
7.	+2000					+2000						
8.												
	4200 + 3800 + 500 + 8000				=	2000 + 3270 + 8800 - 700 + 7500 - 4370			+ 270 + 270			
	16,500					16,500						

$$\therefore \text{Assets} = 4200 + 3800 + 500 + 8000 = \$16,500$$

$$\therefore \text{Liabilities} + \text{Owner's Equity} = 2000 + 3270 + 8800 - 700 + 7500 - 4370 = \$16,500$$

$$\therefore \text{Assets} = \text{Liabilities} + \text{Owner's Equity}$$

(3)

(b)

∴ Income Statement:-

Sonya Jared's Law Office Income Statement For the month of August, 2020		
Revenues		
Service Revenue		\$ 7,500
Expenses		
Salaries Expenses	\$2800	
Rent Expense	900	
Advertising Expense	400	
Utilities Expense	<u>270</u>	
Total expenses		<u>\$ 4370</u>
Net income		<u><u>\$ 3130</u></u>

∴ Owner's Equity Statement:-

Sonya Jared's Law Office Owner's Equity Statement For the month of August, 2020		
Owner's capital August 1		\$8800
Add: Net Income	\$3130	<u>\$ 3130</u>
		\$11930
Less: Drawings		<u>\$ 700</u>
Owner's Capital August 31		<u><u>\$11,230</u></u>

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∴ Balance Sheet :-

Sonya Jared's Law office Balance Sheet August 31, 2020	
<u>Assets</u>	
Cash	\$4200
Accounts Receivable	\$3800
Supplies	\$ 500
Equipment	\$ 8000
Total Assets	<u>\$16,500</u>
<u>Liabilities and Owner's Equity</u>	
Liabilities	
Notes Payable	\$2000
Accounts Payable	\$3270
Total liabilities	<u>\$5270</u>
Owner's Equity	
Owner's capital	<u>\$11,230</u>
	<u>\$16,500</u>

— x —

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Chapter: 02E 2.9

General Journal:-

General Journal				
Date	Account Titles and Explanation	Ref.	Debit(\$)	Credit(\$)
Apr 1.	Cash owner's capital		24,000	24,000
2.	Cash Notes Payable		7000	7000
3	Equipment cash		11000	11000
4.	Rent Expense cash		1200	1200
5.	Supplies Cash		1450	1450
6.	Advertising Expense Accounts Payable		600	600
7	cash Accounts Receivable Service Revenue		2000 16000	18000
8.	Owner's Drawing cash		400	400
9.	Utilities Expense cash		2000	2000
10	Account Payable cash		600	600
11.	Interest Expense Cash		40	40

(6)

12.	Salaries and Wages Expense Cash	6400	6400
23.	Cash Accounts Receivable	12000	12000

— x —

E 2.17(a)

General Journal :-

General Journal				
Date	Accounts Title and Explanation	Ref.	Debit(\$)	Credit(\$)
Oct 1.	Cash Owner's Capital		66,000	66,000
4.	Rent Expense Cash		2,000	2,000
7.	Equipment Cash Accounts Payable		18,000	4,000 14,000
8.	Advertising Expense Cash		500	500
10.	Repairs Expense Accounts Payable		390	390
12.	Accounts Receivable Service Revenue		3,200	3,200
16.	Supplies Accounts Payable		410	410
21.	Accounts Payable Cash		24,000	24,000

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Oct 24.	Utilities Expense		148	
	Cash			148
27.	Cash		3200	
	Accounts Receivable			3200
31.	Salaries and Wages Expense		5100	
	Cash			5100

— x —

(6)

T-accounts :-

Cash	
Debit	Credit
66,000	2000
3200	4000
	500
	14000
	148
	5100

∴ Balance = 43,452

Accounts Receivable	
Debit	Credit
3200	3200
Balance = 0	

(8)

Supplies

Debit	Credit
410	420

Balance = 410

Equipment

Debit	Credit
18000	

∴ Balance = 18000

Accounts Payable

Debit	Credit
14000	14000
	390
	410

∴ Balance = 800

Owner's Capital

Debit	Credit
	66000

∴ Balance = 66000

Service Revenue

Debit	Credit
	3200

∴ Balance = 3200

Rent Expense

Debit	Credit
2000	

∴ Balance = 2000

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Advertising Expense

Debit	Credit
500	

∴ Balance = 500

Repairs Expense

Debit	Credit
390	

∴ Balance = 390

Utilities Expense

Debit	Credit
148	

∴ Balance = 148

Salaries and Wages Expense

Debit	Credit
5100	

∴ Balance = 5100

— x —

(10)

(c)

Trial Balance:-

Bayeru Security Company		
Trial Balance		
October 31, 2020		
	Debit(\$)	Credit(\$)
Cash	43452	
Accounts Receivable	-	
Supplies	420	
Equipment	18000	
Accounts Payable		800
Owner's Capital		66,000
Service Revenue		3200
Rent Expense	2000	
Advertising Expense	500	
Repairs Expense	390	
Utilities Expense	148	
Salaries and Wages Expense	5100	
	<u>70,000</u>	<u>70,000</u>

— x —

P2.1A

1. General Journal:-

General Journal				
Date	Accounts Title and Explanation	Ref.	Debit(\$)	Credit(\$)
Mar 1.	Cash owner's Capital		20000	20000
3.	Land Buildings Equipment cash		12000 2000 1000	15000
5.	Advertising Expense cash		900	900
6.	Prepaid Insurance cash		600	600
10.	Equipment Accounts Payable		1050	1050
18.	Cash Service Revenue		1100	1100
19	Cash Unearned Service Revenue		1500	1500
25	Owner's Drawings cash		800	800
30	Salaries and Wages Expense cash		250	250
30.	Accounts Payable / cash		1050	1050
31.	Cash Service Revenue		2700	2700

P2.2A(a)

∴ General Journal:-

General Journal				
Date	Accounts Title and Explanation	Ref.	Debit(\$)	Credit(\$)
Apr 1	Cash	101	20000	
	Owner's Capital	301		2000
2	Rent Expense	729	1500	
	Cash	101		1500
3	Supplies	126	4000	
	Accounts Payable	201		4000
10	Accounts Receivable	112	5100	
	Service Revenue	400		5100
11	Cash	101	1000	
	Unearned Service Revenue	209		1000
20	Cash	101	2100	
	Service Revenue	400		2100
30	Salaries and Wages Expense	726	2800	
	Cash	101		2800
30	Accounts Payable	201	2600	
	Cash	101		2600

— x —

(b)

Posting to Ledger (T-accounts): -

Cash (101)

Debit	Credit
20000	1500
1000	2800
2100	2600

∴ Balance = 26,200

(13)

Accounts Receivable (112)

Debit	Credit
5100	

∴ Balance = 5100

Supplies (126)

Debit	Credit
4000	

∴ Balance = 4000

Accounts Payable (201)

Debit	Credit
2600	4000

∴ Balance = 1400

Unearned Service Revenue (209)

Debit	Credit
	1000

∴ Balance = 1000

Owner's Capital (301)

Debit	Credit
	20000

∴ Balance = 20000

Service Revenue (400)

Debit	Credit
	5100
	2100

∴ Balance = ~~7200~~ 7200

(14)

Salaries and Wages Expense (726)

Debit

2800

Credit

∴ Balance = 2800

Rent Expense (729)

Debit

1500

Credit

∴ Balance = 1500

— x —

(c)

∴ Trial Balance:-

Vera Ernst
Trial Balance
April 30, 2020

		Debit(\$)	Credit(\$)
101	Cash	16200	
112	Accounts Receivable	5100	
126	Supplies	4000	
201	Accounts Payable		1400
209	Unearned Service Revenue		1000
301	Owner's Capital		20000
400	Service Revenue		7200
726	Salaries and Wages Expense	2800	
729	Rent Expense	1500	
	Totals	<u>29600</u>	<u>29600</u>

— x —

P23A(a)

∴ General Journal: -

General Journal				
Date	Accounts Title and Explanation	Ref.	Debit(\$)	Credit(\$)
May 1.	Cash owner's Capital		40000	40000
1.	Prepaid Rent Cash		24000	24000
1.	Furniture and Equipment Cash Accounts Payable		30000	10000 20000
1.	Prepaid Insurance Cash		1800	1800
6.	Office Supplies Cash		420	420
7.	Office Supplies Accounts Payable		1500	1500
8.	Cash Accounts Receivable Service Revenue		8000 12000	20000
9.	Accounts Payable Cash		400	400
10	Cash Accounts Receivable		3000	3000
11.	Utilities Expense Accounts Payable		380	380
12.	Salaries and Wages Expense Cash		6100	6100

(16)

(b)

T-Accounts:-

Cash	
Debit	Credit
40000	24000
8000	10000
3000	1800
	420
	400
	6100

∴ Balance = 8280

Accounts Receivable	
Debit	Credit
22000	3000

∴ Balance = 9000

Office Supplies	
Debit	Credit
420	
1500	

∴ Balance = 1920

Prepaid Rent	
Debit	Credit
24000	

∴ Balance = 24000

Prepaid Insurance	
Debit	Credit
1800	

∴ Balance = 1800

(17)

Furniture and Equipment

Debit	Credit
30000	

∴ Balance = 30000

Accounts Payable

Debit	Credit
400	20000
	1500
	380

∴ Balance = 21480

Owner's Capital

Debit	Credit
	40000

∴ Balance = 40000

Service Revenue

Debit	Credit
	20000

∴ Balance = 20000

Utilities Expense

Debit	Credit
380	

∴ Balance = 380

Salaries and Wages Expense

Debit	Credit
6100	

∴ Balance = 6100

(18)

(c)

∴ Trial Balance:-

Maquoketa Service
Trial Balance
May 31, 2020

	<u>Debit(\$)</u>	<u>Credit(\$)</u>
Cash	8280	
Accounts Receivable	9000	
Office Supplies	1920	
Prepaid Rent	24000	
Prepaid Insurance	1800	
Furniture and Equipment	30000	
Accounts Payable		21480
Owner's Capital		40000
Service Revenue		20000
Utilities Expense	380	
Salaries and Wages Expense	6100	
	<u>82,480</u>	<u>82,480</u>

— x —

(15)

Chapter: 05

Exercise: 5-4

(1)

Given, Sales = \$200,000

Variable expense = \$120,000

$$\begin{aligned}\therefore \text{Contribution Margin} &= \text{Sales} - \text{Variable Expense} \\ &= \$200,000 - \$120,000 \\ &= \$80,000\end{aligned}$$

$$\begin{aligned}\therefore \text{CM Ratio} &= \frac{\text{Contribution Margin}}{\text{Sales}} \\ &= \frac{80,000}{200,000} \\ &= 0.4 = 40\% \\ &\quad \text{(Answer)}\end{aligned}$$

(2)

$$\text{CM Ratio} = 40\% = 0.4$$

Change in sales = increase of \$1000 = \$1000

$$\begin{aligned}\therefore \text{Change in net operating income} &= \text{change in sales} \times \text{CM ratio} \\ &= 1000 \times 0.4 \\ &= \$400 \\ &\quad \text{(Answer)}\end{aligned}$$

— x —

(20)

Exercise: 5-6

(1)

Given,

Sales per unit = \$15

Variable expense per unit = \$12

∴ CM per unit = \$15 - \$12 = \$3

$$\begin{aligned}\therefore \text{Break-even point in unit sales} &= \frac{\text{Fixed expense}}{\text{CM per unit}} \\ &= \frac{4200}{3} \\ &= 1400 \text{ unit} \\ &\quad (\text{Answer})\end{aligned}$$

(2)

from (1),

CM per unit = \$3

∴ CM ratio = $\frac{3}{15} = 0.2$

$$\begin{aligned}\therefore \text{Break-even point in dollar} &= \frac{\text{Fixed expense}}{\text{CM ratio}} \\ &= \frac{4200}{0.2} \\ &= \$21,000 \\ &\quad (\text{Answer})\end{aligned}$$

~~see above~~

(3)

∴ New fixed expense = \$4200 + \$600 = \$4800

∴ CM per unit = \$3, CM ratio = 0.2

$$\therefore \text{New break-even point in unit} = \frac{4800}{3} = 1600 \text{ units} \quad (\text{Answer})$$

$$\begin{aligned}\therefore \text{New break-even point in dollar} &= \frac{4800}{0.2} \\ &= \$24,000 \quad (\text{Answer})\end{aligned}$$

(21)

Exercise: 5-7

(1)

Let be,
unit = x

$\therefore \text{Sales} = \text{Variable cost} + \text{fixed expense} + \text{profit}$

$$\Rightarrow 120x = 80x + 50000 + 10000$$

$$\therefore x = 1500 \text{ units (Answer)}$$

(2)

Here,
profit = \$15000

$\therefore \text{Sales} = \text{Variable cost} + \text{fixed expense} + \text{profit}$

$$\Rightarrow 120x = 80x + 50,000 + 15000$$

$$\therefore x = 1625 \text{ units}$$

$\therefore$ Dollar sales needed to gain profit \$15000

$$\text{is} = 1625 \times 120 = \$195,000 \text{ (Answer)}$$

— x —

Exercise: 5-8

(1)

Given,

Selling price per unit = \$30

Variable expense per unit = \$20

$$\therefore \text{CM in units} = \$30 - \$20 = \$10$$

$$\therefore \text{CM ratio} = \frac{10}{30} = 0.3333$$

(22)

$$\begin{aligned}\therefore \text{Break-even point in dollars} &= \frac{\text{Fixed Expenses}}{\text{CM Ratio}} \\ &= \frac{7,500}{0.3333} \\ &= \$22,500\end{aligned}$$

Here, unit sales per month = 2000

$$\therefore \text{Actual Sales} = 2000 \times \$30 = \$30,000$$

$$\begin{aligned}\therefore \text{Margin of safety} &= \text{Actual sales} - \text{break even point sales} \\ &= \$30,000 - \$22,500 \\ &= \$7,500 \text{ (Answer)}\end{aligned}$$

$$\begin{aligned}\therefore \text{Margin of safety as percentage} &= \frac{7,500}{30,000} \times 100\% \\ &= 25\% \text{ (Answer)}\end{aligned}$$

Exercise: 5-11

(a)

(a) Case 1

Given,

Units sold = 25,000

Sales = \$180,000

Variable expense = \$120,000

Fixed expenses = \$50,000

$\therefore$ CM per unit = ?

$\therefore$ Net operating income = ?

(23)

$$\therefore \text{CM} = 180,000 - 120,000 = \$60,000$$

$$\therefore \text{CM per unit} = \frac{60,000}{15,000} = 4 \quad (\text{Answer})$$

$$\begin{aligned} \therefore \text{Net operating income} &= \text{CM} - \text{Fixed expenses} \\ &= 60,000 - 50,000 \\ &= \$10,000 \quad (\text{Answer}) \end{aligned}$$

(a) Case 2

Given,

Units sold = ?

Variable expenses = ?

$$\text{Sales} = \$100,000$$

$$\text{CM per unit} = 10$$

$$\text{Fixed expenses} = \$32,000$$

$$\text{Net operating income} = \$8,000$$

$$\therefore \text{Net operating income} = \text{CM} - \text{Fixed expenses}$$

$$\Rightarrow 8000 = \text{CM} - 32000$$

$$\therefore \text{CM} = 40,000$$

$$\therefore \text{Units Sold} = \frac{\text{CM}}{\text{CM per unit}}$$

$$= \frac{40,000}{10}$$

$$= 4,000 \text{ Answer}$$

$$\therefore \text{CM} = \text{Sales} - \text{Variable expense}$$

$$\Rightarrow 40,000 = 100,000 - \text{variable expenses}$$

$$\therefore \text{Variable expenses} = \$60,000 \quad (\text{Answer})$$

(24)

(a) Case 3

Given,

$$\text{Units sold} = 10,000$$

$$\text{Variable expenses} = \$70,000$$

$$\text{CM per unit} = \$13$$

$$\text{Net operating income} = \$12,000$$

$$\therefore \text{Sales} = ?$$

$$\therefore \text{Fixed expenses} = ?$$

$$\therefore \text{Units sold} = \frac{\text{CM}}{\text{cm per unit}}$$

$$\Rightarrow 10,000 = \frac{\text{CM}}{13}$$

$$\therefore \text{CM} = \$130,000$$

$$\therefore \text{CM} = \text{Sales} - \text{Variable expenses}$$

$$\Rightarrow 130,000 = \text{Sales} - 70,000$$

$$\therefore \text{Sales} = \$200,000 \text{ (Answer)}$$

$$\therefore \text{Net operating income} = \text{CM} - \text{Fixed Expenses}$$

$$\Rightarrow 12,000 = 130,000 - \text{Fixed expenses}$$

$$\therefore \text{Fixed expenses} = \$118,000 \text{ (Answer)}$$

(a) Case 4

Given,

$$\text{Units sold} = 6000$$

$$\text{Sales} = \$300,000$$

$$\text{Fixed expenses} = \$100,000$$

(25)

Net operating income (loss) = \$(10,000)

Net operating income = CM - Fixed expenses

$$\Rightarrow -10,000 = \text{CM} - 100,000$$

$$\therefore \text{CM} = \$90,000$$

$\therefore \text{CM} = \text{Sales} - \text{Variable expense}$

$$\Rightarrow 90,000 = 300,000 - \text{Variable expense}$$

$\therefore \text{Variable expenses} = \$210,000$ (Answer)

$$\therefore \text{Units sold} = \frac{\text{CM}}{\text{CM per unit}}$$

$$\Rightarrow 6,000 = \frac{\cancel{300,000} \text{ CM}}{\text{CM per unit}} \frac{90,000}{\text{CM per unit}}$$

$$\therefore \text{CM per unit} = 15 \text{ (Answer)}$$

— x —

(b)

(b) Case 1

Given,

$$\text{Sales} = \$500,000$$

$$\text{CM ratio} = 20\%$$

$$\text{Net operating income} = \$7000$$

$$\therefore \text{CM} = \text{CM ratio} \times \text{Sales} = 20\% \times 500,000 = \$100,000$$

$$\therefore \text{Variable expense} = 500,000 - 100,000 = \$400,000$$

(Answer)

(26)

$$\therefore \text{Fixed expense} = 100,000 - 7000 = \$93,000 \quad (\text{Answer})$$

(b) Case 2

Given,

$$\text{Sales} = \$400,000$$

$$\text{Variable expenses} = \$260,000$$

$$\text{Fixed expenses} = \$100,000$$

$$\therefore \text{CM} = 400,000 - 260,000 = \$140,000$$

$$\therefore \text{CM Ratio} = \frac{140,000}{400,000} = 35\% \quad (\text{Answer})$$

$$\begin{aligned} \therefore \text{Net operating income} &= 140,000 - 100,000 \\ &= \$40,000 \quad (\text{Answer}) \end{aligned}$$

(b) Case 3

Given,

$$\text{CM Ratio} = 60\%$$

$$\text{Fixed expenses} = \$130,000$$

$$\text{Net operating income} = \$20,000$$

$$\therefore \text{CM} = 130,000 + 20,000 = \$150,000$$

$$\therefore \text{CM Ratio} = \frac{\text{CM}}{\text{Sales}}$$

$$\therefore 60\% = \frac{150,000}{\text{Sales}}$$

$$\therefore \text{Sales} = \$250,000 \quad (\text{Answer})$$

$$\therefore \text{Variable expenses} = 250,000 - 150,000 \\ = \$100,000 \quad (\text{Answer})$$

(b) Case 4

Given,

$$\text{Sales} = \$600,000$$

$$\text{Variable expenses} = \$420,000$$

$$\text{Net operating income loss} = \$ (5,000)$$

$$\therefore \text{CM} = 600,000 - 420,000 = \$180,000$$

$$\therefore \text{CM Ratio} = \frac{180,000}{600,000} = 30\% \quad (\text{Answer})$$

$$\therefore \text{Net operating income} = \text{CM} - \text{Fixed expenses}$$

$$\Rightarrow -5,000 = 180,000 - \text{Fixed expenses}$$

$$\therefore \text{Fixed expenses} = \$185,000 \quad (\text{Answer})$$

— x —